

U3-D5 Homework – Rational Expressions

Name: Key

Factor
Simplify completely. State excluded values.

$$1. \frac{3x}{12x - x\sqrt{6}}$$

$$= \frac{\cancel{3} \cdot \cancel{x}}{\cancel{x} (12 - \sqrt{6})}$$

$$= \boxed{\frac{3}{12 - \sqrt{6}}}$$

don't cancel 3 & 12

$$2. \frac{10 - \sqrt{40}}{-6 + \sqrt{-16}}$$

$$= \frac{10 - \sqrt{4 \cdot 10}}{-6 + 4i}$$

$$= \frac{10 - 2\sqrt{10}}{-6 + 4i} \leftarrow \text{Factor}$$

$$= \frac{\cancel{2} (5 - \sqrt{10})}{\cancel{2} (-3 + 2i)} = \boxed{\frac{5 - \sqrt{10}}{-3 + 2i}}$$

$$3. \frac{x^2 - 2x + 1}{x^2 - 1} \leftarrow a^2 - b^2$$

if there's an "x" in denominator,
there will be excluded values

$$\frac{(\cancel{x} - 1)(x - 1)}{(x + 1)(\cancel{x} - 1)}$$

$$\downarrow$$

$$\boxed{\frac{(x - 1)}{(x + 1)}}$$

excluded values

$$x \neq 1, -1$$

$$4. \frac{-3x^2 - 15x}{x^2 - 25} \leftarrow a^2 - b^2$$

$$= \frac{-3x(\cancel{x} + 5)}{(\cancel{x} + 5)(x - 5)}$$

$$= \boxed{\frac{-3x}{(x - 5)}}$$

excluded values

$$x \neq 5, -5$$

$$5. \frac{x^2 + 13x + 40}{x^3 - 64x}$$

$$= \frac{(x + 5)(x + 8)}{x(x^2 - 64)} \leftarrow a^2 - b^2$$

$$= \frac{(x + 5)(\cancel{x} + 8)}{x(\cancel{x} + 8)(x - 8)}$$

$$\boxed{\frac{(x + 5)}{x(x - 8)}}$$

excluded values

$$x \neq 0, 8, -8$$

$$6. \frac{4x^2 - 12x + 9}{6x^2 - 19x + 15}$$

$$= \frac{4x^2 - 6x - 6x + 9}{6x^2 - 9x - 10x + 15}$$

$$= \frac{2x(2x - 3) - 3(2x - 3)}{3x(2x - 3) - 5(2x - 3)}$$

$$= \frac{(\cancel{2x} - 3)(2x - 3)}{(3x - 5)(\cancel{2x} - 3)}$$

$$= \boxed{\frac{(2x - 3)}{(3x - 5)}}$$

excluded values:
 $x \neq 5/3, 3/2$

(Set factors $\neq 0$)

$$\begin{array}{ll} 3x - 5 \neq 0 & 2x - 3 \neq 0 \\ x \neq 5/3 & x \neq 3/2 \end{array}$$

Product/Sum factoring

$$\frac{36}{-6 \cdot -6}$$

$$\frac{90}{-9 \cdot -10}$$

video on google classroom

Feel free to search online for "polysmlt" or "polysmlt quadratic" to find instructions on using an app built into the TI-84 that you can use to solve quadratics automatically. It's less hassle than entering the quadratic formula.

A geyser launches a pinecone into the air at a speed of 66m/s. The bottom of the geyser is at the top of a 12-meter hill.

$$h(t) = -4.9t^2 + V_0 t + h_0$$

V_0 = initial speed
 h_0 = initial height

7. Write functions to describe both the height and the velocity of the pinecone after t seconds.

$$h(t) = -4.9t^2 + 66t + 12$$

8. What is the height of the pinecone after 5 seconds?

$$\begin{aligned} h(5) &= -4.9(5)^2 + 66(5) + 12 \\ &= 219.5 \text{ meters} \end{aligned}$$

9. When will the pinecone be 140 feet high?

$$140 = -4.9t^2 + 66t + 12$$

$$0 = -4.9t^2 + 66t - 128 \quad \leftarrow \text{use POLYSMLT}$$

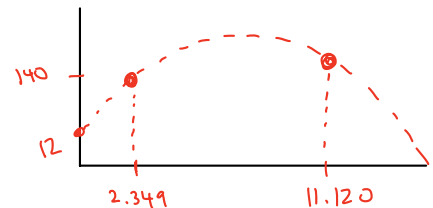
$$\begin{aligned} t &= 2.349 \\ t &= 11.120 \text{ seconds} \end{aligned}$$

10. How fast is it moving at each of the times from #9?

$$V(t) = at + V_0 \quad a = -9.8 \text{ m/s}^2 \quad \leftarrow \text{gravity}$$

$$V(2.349) = -9.8(2.349) + 66 = 42.98 \text{ m/s} \quad \leftarrow \text{going up}$$

$$V(11.120) = -9.8(11.120) + 66 = -42.98 \text{ m/s} \quad \leftarrow \text{going down}$$



11. When will the pinecone hit the ground?

$$\text{When } h(t) = 0$$

$$0 = -4.9t^2 + 66t + 12 \quad \leftarrow \text{use POLYSMLT}$$

$$t = -0.179 \quad t = 13.649 \text{ seconds}$$

12. What is its speed on impact?

$$V(13.649) = -9.8(13.649) + 66 = -67.76 \text{ m/s}$$

13. What are the practical domain and range of the height function?

$$\text{Domain: } [0, 13.649]$$

$$\text{Range: (we need height at vertex)} = \frac{-b}{2a} = \frac{-66}{2(-4.9)} = 6.735 \text{ seconds}$$

$$h(6.735) = -4.9(6.735)^2 + 66(6.735) + 12 = 423.509 \text{ m} \quad \leftarrow \text{max height}$$

$$\text{Range: } [0, 234.245]$$